

Team Details

- Team name: **BYTE ME**
- Team leader name: **Jyotier Vithalani**
- Team leader email-id: **Jyotier.v@ahduni.edu.in**
- Problem Statement: **Cyber-Resilient Infrastructure for Critical Sectors**
- Team Members: **Tanmay Kanani, Krishna Patel, Jinay Shah, Akshay Patel**

Problem Overview

Critical digital infrastructures in healthcare, agriculture, and urban systems operate at large scale and are expected to be always available and secure. However, these **systems are increasingly exposed to sophisticated cyber threats**, many of which operate over encrypted traffic and evade traditional security mechanisms.

Our Idea

- An intelligent, behaviour-based security system that **detects zero-day and advanced threats hidden in encrypted traffic without decryption**.
- **Machine learning-driven** anomaly detection enables automated, **rapid threat containment** and delivers an adaptive, **future-ready cyber-resilience layer** for critical infrastructure.



Our Progress

- Implemented kuberneteses with horizontal pod scaling.
- Add email on alert/attack functionality.
- Added rate limiting (customisable value) in the node software.
- Added ip blocking in by node.
- Developed a unified setup utility that provisions the complete Kubernetes environment.
- Federated learning on telemetry data.

Proof of Concept

📄 Problem → Solution Mapping		
Problem Requirement	Your Solution	Status
Real-time monitoring	Ingest Service + WebSocket	✓
Anomaly detection algorithms	ML Models (LSTM, RF) + Rule Engine	✓
Alerting system with severity levels	Alert Manager (low/medium/high/critical)	✓
Automated response mechanisms	Response Engine + Playbooks	✓
Dashboard showing security status	React Frontend	✓
Simulated attack scenarios	Attack Simulator Scripts	✓
Healthcare/Agriculture/Urban domains	Domain-specific ML models	✓
Scalability	Kubernetes + HPA	✓



Approach

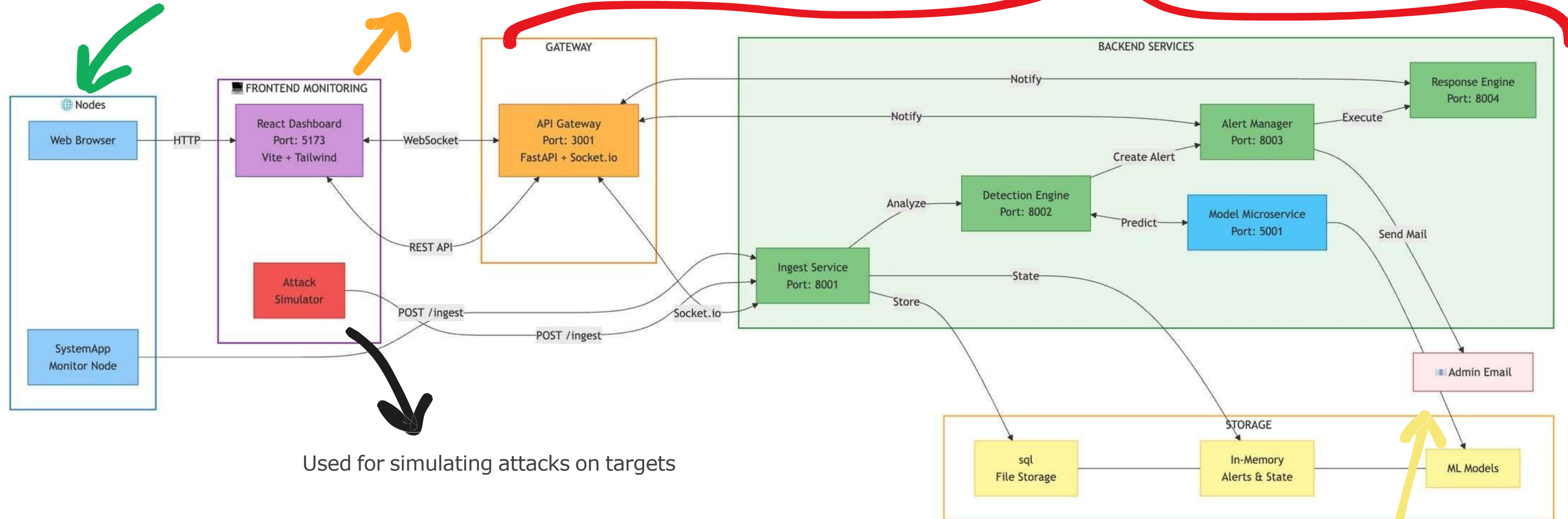
- Traffic Capture & Flow Construction
- **Network traffic is captured at the edge**, extracting timing, volume, and protocol-level metadata.
- Encryption-Aware Routing
- **Rule-Based Analysis for Unencrypted Traffic**
- Unencrypted traffic is inspected using predefined signatures and security rules for fast, deterministic detection.
- Behaviour-Based Analysis for Encrypted Traffic
- **Encrypted traffic bypasses payload inspection and is analysed using flow-level behavioural features.**
- Domain-Aware Model Selection
- The system dynamically selects the appropriate detection model based on the operational context:
 - General Network → **Neural Network Classifier**
 - Agriculture → **LSTM Autoencoder**
 - Healthcare → **LSTM Classifier**
 - Urban → **LSTM Forecaster**
- Threat Scoring & **Decision Engine**
- Automated & Manual Response Execution
- **Federated Learning Feedback Loop**
- **Local learnings are aggregated centrally and redistributed,**

Architecture Diagram

Each service has its own **docker** image which is then scaled on **kubernetees**

Node side software

Monitoring dashboard



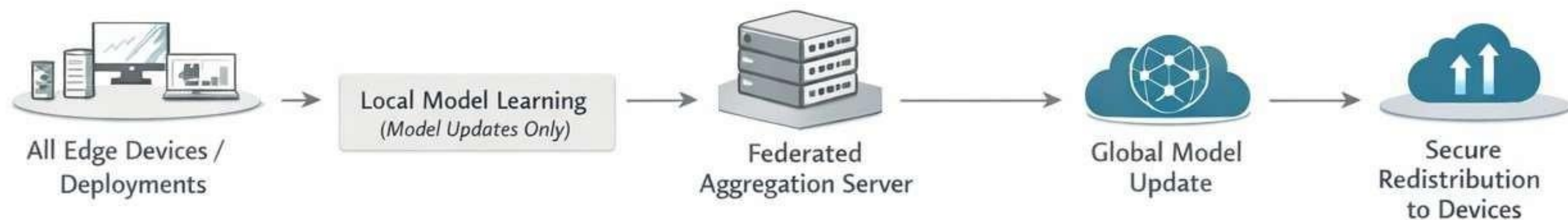
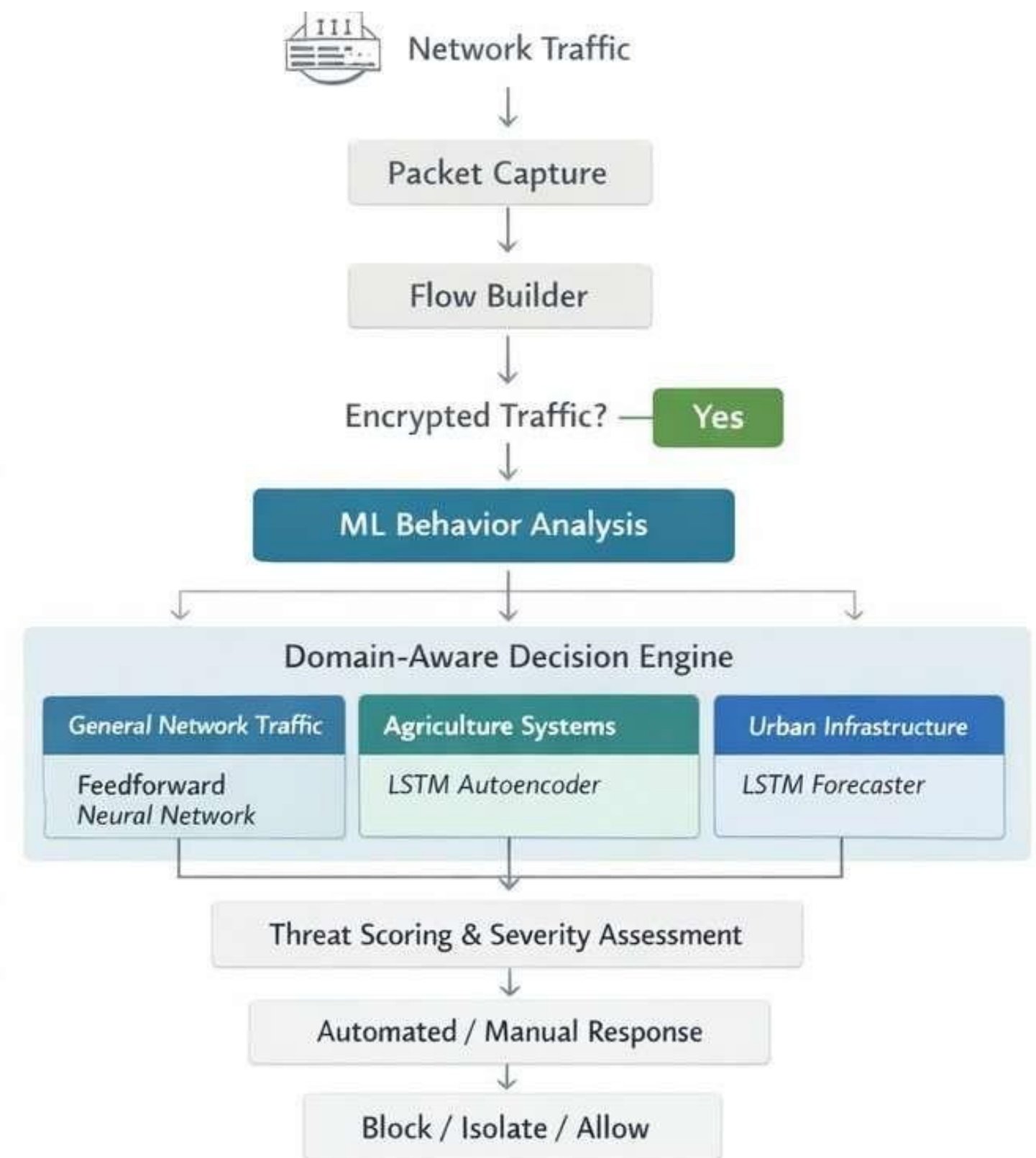
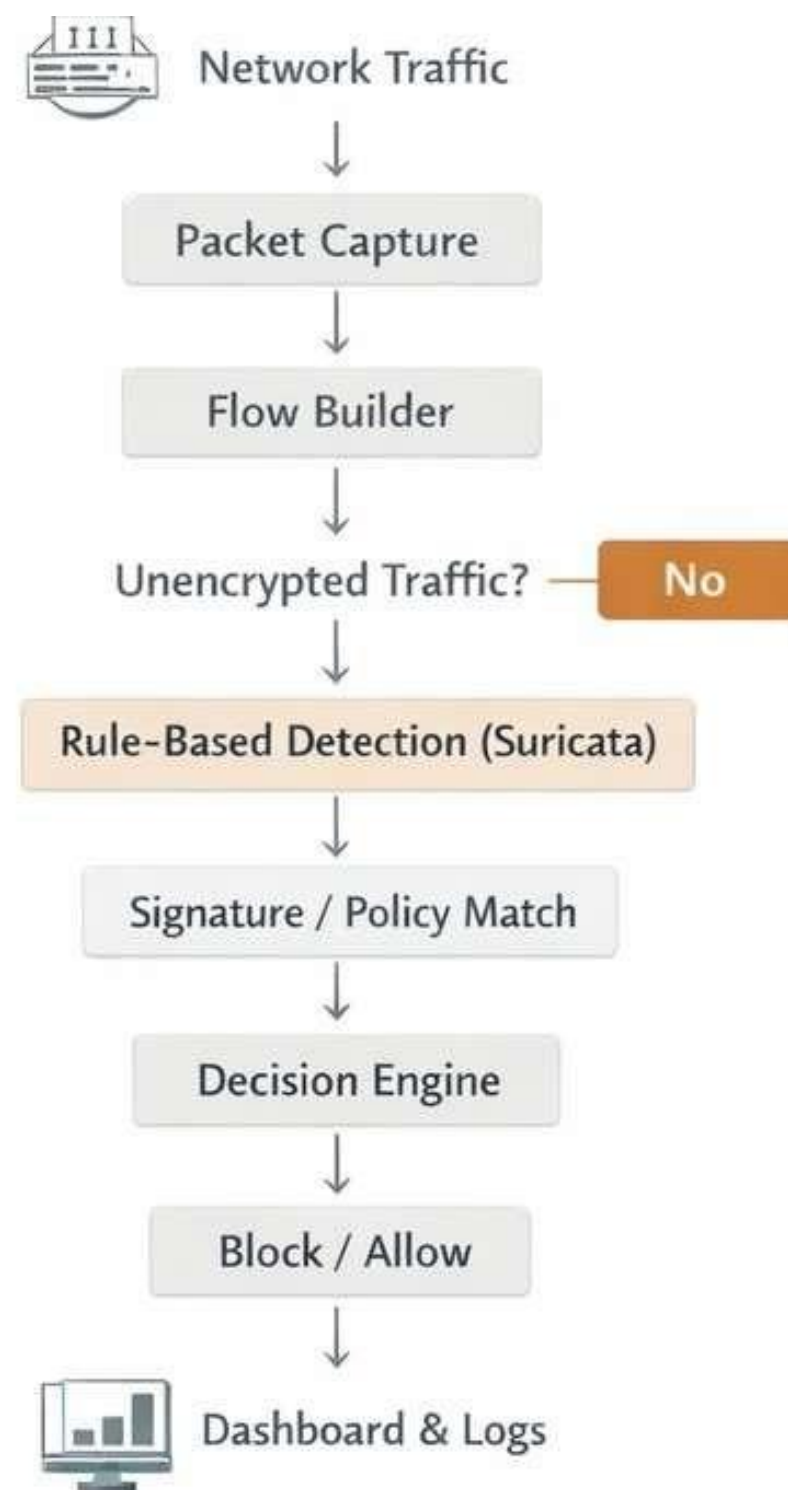
Used for simulating attacks on targets

Email notification

Machine Learning Models Used

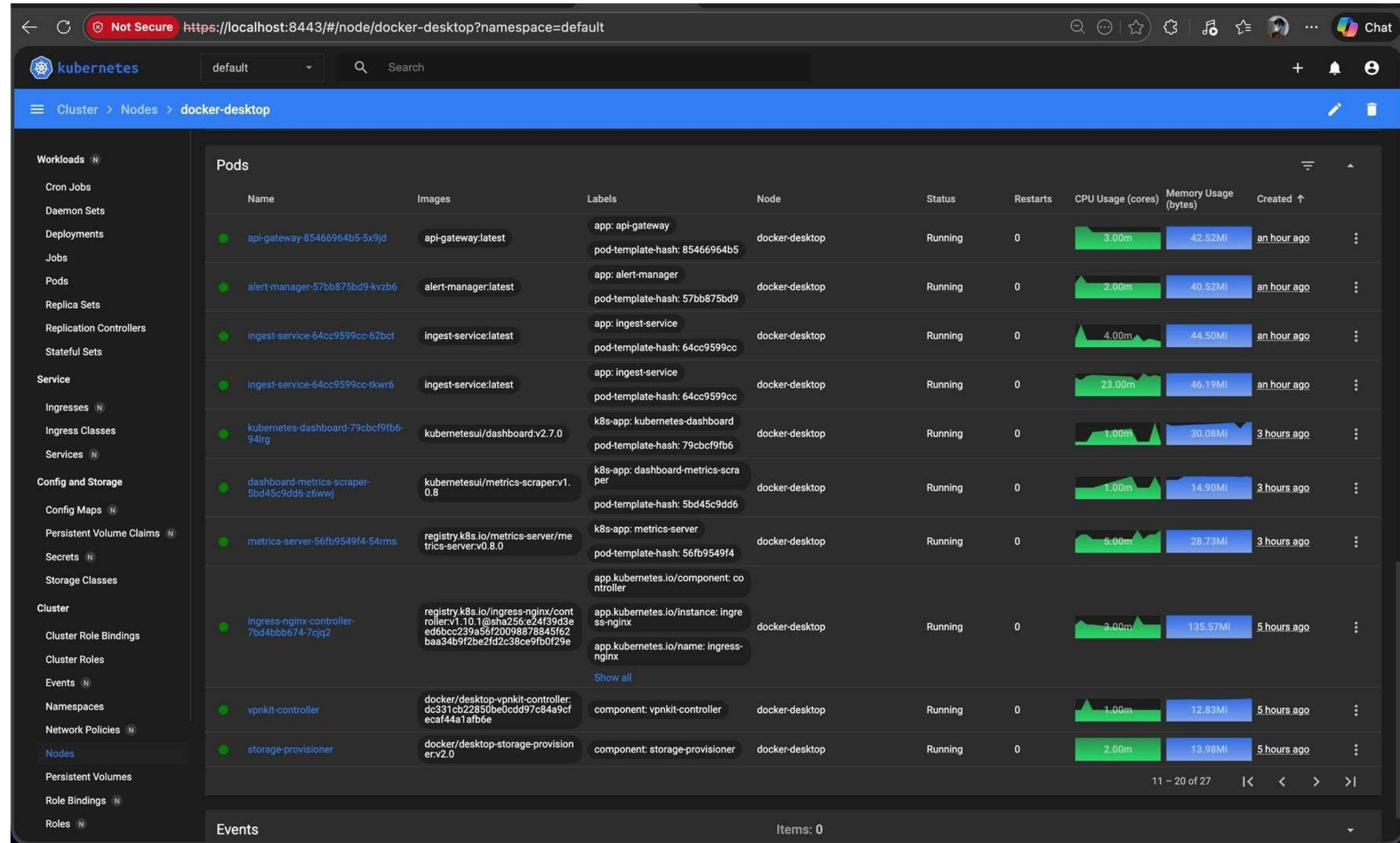
1. **Feedforward Neural Network** (PyTorch): **Generic network attack detection** (DDoS, Botnet, Brute Force)
2. **Isolation Forest** (Unsupervised ML): Network flow based anomaly detection on encrypted traffic
3. **LSTM Autoencoder** (Unsupervised): Agriculture sensor spoofing and physics-based anomaly detection
4. **LSTM Classifier** (Supervised): Healthcare IoMT DDoS and traffic surge detection
5. **LSTM Forecaster** (Regression): Urban traffic flow prediction and grid manipulation detection

Detection engine



Scalability : Kubernetes

- The platform uses Kubernetes **Horizontal Pod Autoscaler** to dynamically scale services based on real-time resource utilization.
- **Self-Healing Infrastructure:** Failed or compromised pods are automatically recreated.
- **Multi-Replica Critical Services:** Detection, alerting, and dashboard services run with replicas.



The screenshot displays the Kubernetes dashboard interface in a web browser. The URL is `https://localhost:8443/#/node/docker-desktop?namespace=default`. The dashboard shows a list of pods running on the 'docker-desktop' node. The pods are organized into a table with columns for Name, Images, Labels, Node, Status, Restarts, CPU Usage (cores), Memory Usage (bytes), and Created. The pods are all in a 'Running' state. The dashboard also includes a sidebar with navigation links for Workloads, Service, Config and Storage, Cluster, Namespaces, Network Policies, Persistent Volumes, Role Bindings, and Roles. The 'Pods' section is currently selected, showing a list of 11 pods. The pods are: api-gateway-85466964b5-5x9jd, alert-manager-57bb875bd9-kvzb6, ingest-service-64cc9599cc-62bct, ingest-service-64cc9599cc-1kwr6, kubernetes-dashboard-79cbcf9fb6-94lrg, dashboard-metrics-scraper-5bd45c9dd6-z6wwj, metrics-server-56fb9549f4-54rms, ingress-nginx-controller-7bd4bbb674-7cjq2, vpnkit-controller, and storage-provisioner. The dashboard also shows a 'Events' section at the bottom with 0 items.

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
api-gateway-85466964b5-5x9jd	api-gateway:latest	app: api-gateway pod-template-hash: 85466964b5	docker-desktop	Running	0	3.00m	42.52Mi	an hour ago
alert-manager-57bb875bd9-kvzb6	alert-manager:latest	app: alert-manager pod-template-hash: 57bb875bd9	docker-desktop	Running	0	2.00m	40.52Mi	an hour ago
ingest-service-64cc9599cc-62bct	ingest-service:latest	app: ingest-service pod-template-hash: 64cc9599cc	docker-desktop	Running	0	4.00m	44.50Mi	an hour ago
ingest-service-64cc9599cc-1kwr6	ingest-service:latest	app: ingest-service pod-template-hash: 64cc9599cc	docker-desktop	Running	0	23.00m	46.19Mi	an hour ago
kubernetes-dashboard-79cbcf9fb6-94lrg	kubernetesui/dashboard:v2.7.0	k8s-app: kubernetes-dashboard pod-template-hash: 79cbcf9fb6	docker-desktop	Running	0	1.00m	30.08Mi	3 hours ago
dashboard-metrics-scraper-5bd45c9dd6-z6wwj	kubernetesui/metrics-scraper:v1.0.8	k8s-app: dashboard-metrics-scraper pod-template-hash: 5bd45c9dd6	docker-desktop	Running	0	1.00m	14.90Mi	3 hours ago
metrics-server-56fb9549f4-54rms	registry.k8s.io/metrics-server/metrics-server:v0.8.0	k8s-app: metrics-server pod-template-hash: 56fb9549f4	docker-desktop	Running	0	5.00m	28.73Mi	3 hours ago
ingress-nginx-controller-7bd4bbb674-7cjq2	registry.k8s.io/ingress-nginx/controller-v1.10.1@sha256:e24f39d3e6d6bce239a56f20098878845f62baa34b9f2be2fd2c38ce9fb0f29e	app.kubernetes.io/component: controller app.kubernetes.io/instance: ingress-nginx app.kubernetes.io/name: ingress-nginx	docker-desktop	Running	0	3.00m	135.57Mi	5 hours ago
vpnkit-controller	docker/desktop-vpnkit-controller:dc331cb22850be0cdd97c84a9cfecaf44a1afb6e	component: vpnkit-controller	docker-desktop	Running	0	1.00m	12.83Mi	5 hours ago
storage-provisioner	docker/desktop-storage-provisioner:v2.0	component: storage-provisioner	docker-desktop	Running	0	2.00m	13.98Mi	5 hours ago

```
(base) jay@Jays-MacBook-Air Smooth_Operator % ./k8s/kuber_start.sh
Starting Threat Ops Kubernetes Deployment...
✓ Kubernetes cluster detected
✓ Docker images found
Deploying to Kubernetes...
deployment.apps/ingest-service configured
service/ingest-service unchanged
deployment.apps/api-gateway configured
service/api-gateway unchanged
deployment.apps/detection-engine configured
service/detection-engine unchanged
deployment.apps/alert-manager configured
service/alert-manager unchanged
deployment.apps/response-engine configured
service/response-engine unchanged
deployment.apps/frontend configured
service/frontend unchanged
deployment.apps/model-microservice configured
service/model-microservice unchanged
deployment.apps/systemapp configured
service/systemapp unchanged
ingress.networking.k8s.io/threat-ops-ingress unchanged
```


ZERo-DAY & EnCRYPtED AttACK DEtECtion

- **Live Packet Capture** : Uses **unsupervised machine learning** to learn effectively detects previously unseen attacks that bypass signature-based security systems

AUTOMATED RESPONSE & POLICY ENFORCEMENT (SOAR)



- Automatically blocks malicious IPs, and escalates alerts based on threat severity and confidence levels.

LAYERED AI DEfEnSE ARCHitECtuRE



- Ensures early-stage filtering of broad attacks while enabling **deep contextual analysis** for healthcare, agriculture, and urban systems.

PRoJECT FEAtuRES

REAL-TIME DASHBoARD & ViSuAlizAtion

- Displays **live telemetry, alert timelines, device status**, and response actions through **WebSockets**.

FEDERAtED LEARninG-EnABLED COLLEctive DEfEnSE

- Only model updates and threat insights are shared and **aggregated centrally** to improve global detection accuracy.

KUBERNETES-BASED SCALABLE DEPLOYMENT

- Uses containerization and Kubernetes orchestration to automatically scale edge services and central components based on traffic load, ensuring high availability, fault tolerance, and consistent performance under attack conditions.

AttACK SimulAtion & TESTinG FRAmEwoRK

- Includes **attacker and victim simulation** tools to validate detection and response workflows.

AutomAtED ALERt GENERAtion & SEvERitY CLASSifiCAtion

- Transforms detected security issues into human-readable alerts through emails with clear severity levels.

CoDE SNiPPeTs

ANN - GEnERiC nEtwoRK AttACKS

```
1 client_id = data.get('client_id', 'unknown')
2 client_weights = data.get('weights')
3
4 if not client_weights:
5     print(f"[ERR] No weights in payload from {client_id}")
6     return jsonify({"error": "Missing weights"}), 400
7
8 print(f"[INFO] Processing update from {client_id} (Keys: {len(client_weights)})")
9
10 client_state_dict = {k: torch.tensor(v).to(DEVICE) for k, v in client_weights.items()}
11
12 pending_updates.append(client_state_dict)
13 print(f"[FL] Received FL update from {client_id}. Progress: {len(pending_updates)}/{REQUIRED_UPDATES_FOR_AVG}")
14
15 if len(pending_updates) >= REQUIRED_UPDATES_FOR_AVG:
16     print("[FL] Aggregating Global Model (FedAvg)...")
17
18     # FEDERATED AVERAGING ALGORITHM
19     avg_weights = copy.deepcopy(GLOBAL_MODEL.state_dict())
20     for key in avg_weights.keys():
21         # Stack all updates for this key and take mean
22         # Ensure all updates have this key
23         valid_updates = [u[key] for u in pending_updates if key in u]
24         if valid_updates:
25             stacked = torch.stack(valid_updates)
26             avg_weights[key] = torch.mean(stacked, dim=0)
27
28     # Update Global Model
29     GLOBAL_MODEL.load_state_dict(avg_weights)
30     FL_ROUND += 1
31     pending_updates = [] # Reset
32     print(f"[SUCCESS] Global Model Updated! New Round: {FL_ROUND}")
```

```
1 @app.route('/api/fl/submit-update', methods=['POST'])
2 def submit_update():
```

FEDERATED LEARNinG EnDPoint

```
1 train_idx, _ = train_test_split(range(len(tensor_x)), test_size=0.2)
2 loader = DataLoader(TensorDataset(tensor_x[train_idx], tensor_y[train_idx]), batch_size=2048, shuffle=True)
3
4 model = GeneralNetworkShield(input_dim=X.shape[1]).to(DEVICE)
5 optimizer = optim.Adam(model.parameters(), lr=0.001)
6 criterion = nn.BCELoss()
7
8 model.train()
9 for epoch in range(3):
10     total_loss = 0
11     for X_batch, y_batch in loader:
12         X_batch, y_batch = X_batch.to(DEVICE), y_batch.to(DEVICE)
13         optimizer.zero_grad()
14         out = model(X_batch)
15         loss = criterion(out, y_batch)
16         loss.backward()
17         optimizer.step()
18         total_loss += loss.item()
19     print(f"    Epoch {epoch+1} Loss: {total_loss/len(loader):.4f}")
```

POST http://127.0.0.1:5000/api/analyze

Body

```
1 {
2   "sector": "healthcare",
3   "payload": "SELECT * FROM patients WHERE id=1 OR 1=1",
4   "sensor_data": [60, 10, 5, 5]
5 }
```

200 OK 70 ms 339 B

Body

```
1 {
2   "message": "Malicious Web Payload Detected (SQLi/XSS)",
3   "source": "Web Gatekeeper",
4   "status": "blocked",
5   "threat_level": "high"
6 }
```

PoStmAn moDEL tEStinG

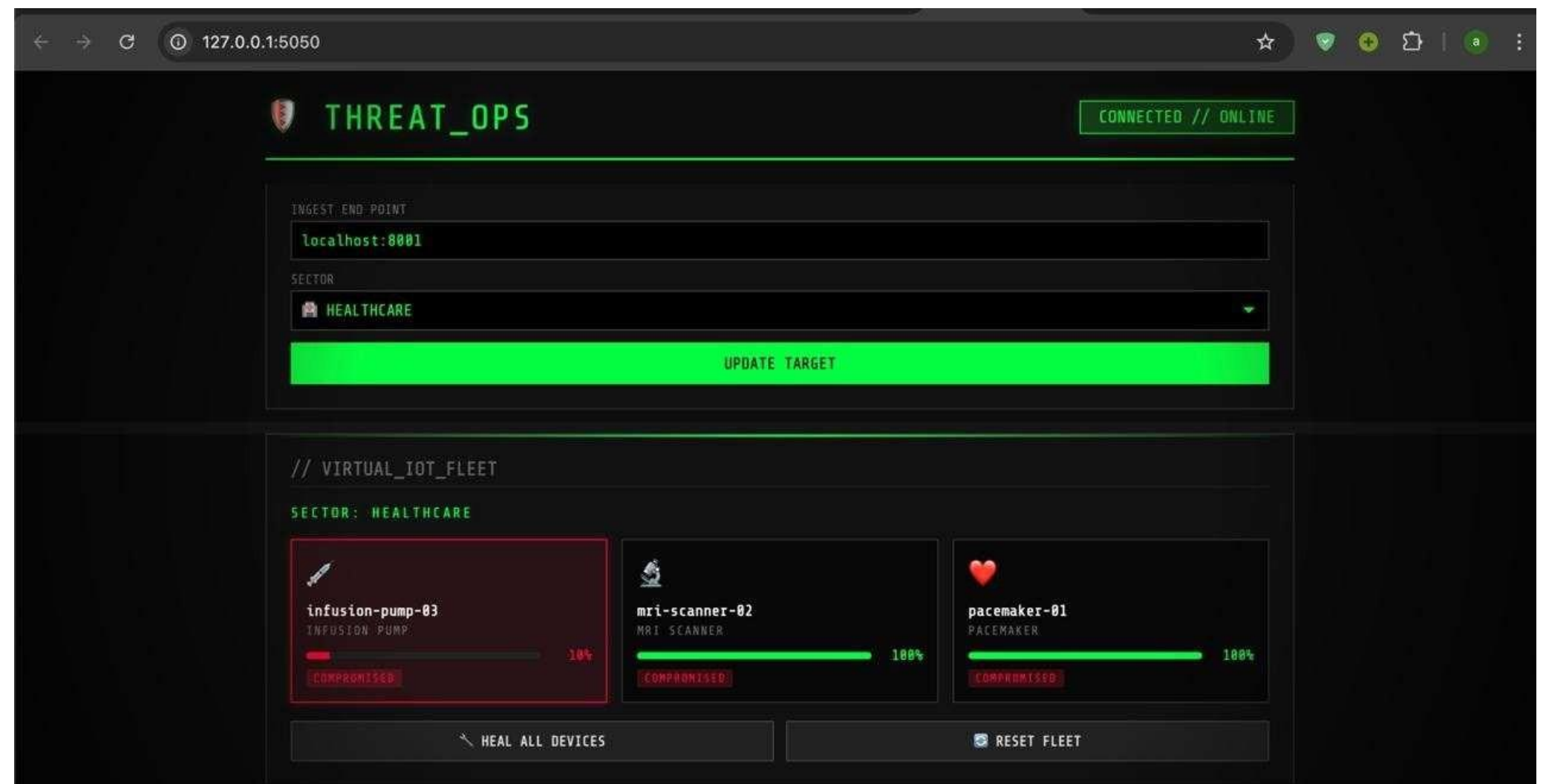
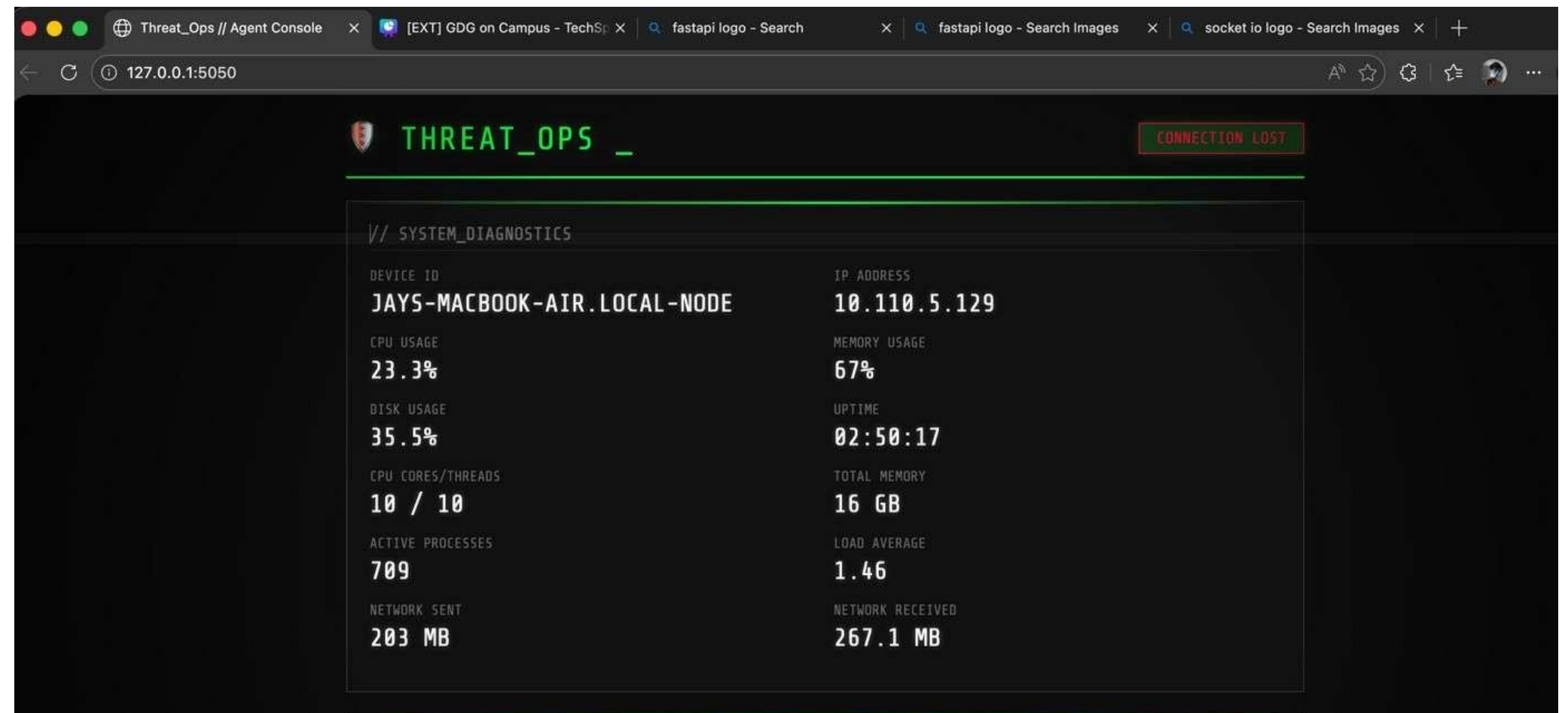
HEALTH CARE ML MODEL

	precision	recall	f1-score	support
Normal	1.00	0.98	0.99	47974
DDoS	0.99	1.00	0.99	51893
accuracy			0.99	99867
macro avg	0.99	0.99	0.99	99867
weighted avg	0.99	0.99	0.99	99867

Accuracy: 0.9910280673295483

Snapshot of Node software

- **Agent** installed on protected machines - runs as a **background daemon**.
- Auto-discovers device identity and sends **real-time telemetry** to our dashboard.
- **Captures attacks** while monitoring **system health**.
- **Cross-platform support** - same agent works on **Windows, Mac, and Linux**
- Supports **headless** systems, using command line arguments.



Snapshot of Monitoring dashboard




```
Windows PowerShell
PS C:\Users\Het Ashishbhai Modi\idk> python .\monitor_server.py

Threat_Ops Universal Agent
Device: Het-Workspace-node
IP: 10.110.5.149
Server: http://10.110.5.98:8001/ingest

Telemetry Agent started. Reporting to http://10.110.5.98:8001/ingest
* Serving Flask app 'monitor_server'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5050
* Running on http://10.110.5.149:5050
Press CTRL+C to quit
10.110.5.98 - - [18/Jan/2026 01:14:31] "OPTIONS /data HTTP/1.1" 200 -
[SECURITY] Query received: SELECT * FROM users WHERE id = 1 OR 1=1
Forwarded alert to Main Server: 200
10.110.5.98 - - [18/Jan/2026 01:14:31] "POST /data HTTP/1.1" 200 -
10.110.5.98 - - [18/Jan/2026 01:14:36] "OPTIONS /data HTTP/1.1" 200 -
[SECURITY] Query received: SELECT * FROM users WHERE id = 1 OR 1=1
Forwarded alert to Main Server: 200
10.110.5.98 - - [18/Jan/2026 01:14:37] "POST /data HTTP/1.1" 200 -
10.110.5.98 - - [18/Jan/2026 01:14:38] "OPTIONS /login HTTP/1.1" 200 -
[SECURITY] Login attempt for user: admin from 10.110.5.98
10.110.5.98 - - [18/Jan/2026 01:14:39] "OPTIONS /login HTTP/1.1" 200 -
[SECURITY] Login attempt for user: admin from 10.110.5.98
Forwarded alert to Main Server: 200
10.110.5.98 - - [18/Jan/2026 01:14:39] "POST /login HTTP/1.1" 401 -
Forwarded alert to Main Server: 200
10.110.5.98 - - [18/Jan/2026 01:14:39] "POST /login HTTP/1.1" 401 -
10.110.5.98 - - [18/Jan/2026 01:14:39] "OPTIONS /login HTTP/1.1" 200 -
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10.110.5.98 - - [18/Jan/2026 01:14:39] "OPTIONS /login HTTP/1.1" 200 -
[SECURITY] Login attempt for user: admin from 10.110.5.98
Forwarded alert to Main Server: 200
10.110.5.98 - - [18/Jan/2026 01:14:39] "POST /login HTTP/1.1" 401 -
10.110.5.98 - - [18/Jan/2026 01:14:40] "OPTIONS /login HTTP/1.1" 200 -
```

Backend server

IronDome .ai
Security Operations Center

ConnectedEvents: 0/minAlerts: 50Blocked: 10011:33:33
Thu, Aug 27, 2026

SearchCtrl K

Tanmay Loq Node
TANMAY-LOQ-nodeOnline

TelemetryTimelineIoT FleetResponseDropped 100

🛡️ DROPPED PACKETS

100 requests blocked

📧 SQL INJECTION100

🔑 BRUTE FORCE0

⚡ FLOODING0

🚫 BLOCKED IP0

🕒 RATE LIMITED0

📄 XSS0

LIVE FEED

50 events

📧 127.0.0.1CRITICAL

SQL INJECTION → /unknown

11:28:53

BLOCKED

📧 127.0.0.1CRITICAL

SQL INJECTION → /unknown

11:28:53

BLOCKED

📧 127.0.0.1CRITICAL

SQL INJECTION → /unknown

11:28:53

BLOCKED

📧 127.0.0.1CRITICAL

SQL INJECTION → /unknown

11:28:53

BLOCKED

🔍 All malicious requests are logged and blocked at the API Gateway

Gateway: localhost:3001

50Total

18Warning

32Critical

⚠️ ML: Network Shield🔍 AI

DDoS/Botnet Activity Detected (High SYN count)

Confidence: 65%

Unknown • 0s ago

✓ Ack✕

🕒 High Network Traffic (Data Exfiltration Risk)

Network traffic spike detected at 2513 KB/s. This could indicate data exfiltration or a flood attack.

TANMAY-LOQ-node • 0s ago

✓ Ack✕

⚠️ Critical Memory Usage

Memory usage at 92.0% on TANMAY-LOQ-node. Critical memory pressure may cause service crashes or system instability.

✓ Ack All✕ Clear Ack'd

🛡️ Run Defense Playbook

1Total Nodes

100%Health

telemetry.stream100 events

11:33:33TANMAY-LOQ-nodeCPU 15.4%MEM 91.8%NET 2513KB/sREQ 0

11:33:32TANMAY-LOQ-nodeCPU 16.1%MEM 91.9%NET 2520KB/sREQ 0

11:33:32TANMAY-LOQ-nodeCPU 15.2%MEM 92.8%NET 2510KB/sREQ 0

11:33:31TANMAY-LOQ-nodeCPU 16.7%MEM 91.8%NET 2520KB/sREQ 0

173

🛡️ Defense Active!

Defense Active! Executed 6 actions on 6 alerts.

✕

⚡ ACTIONS EXECUTED (6)

➡➡ IP 10.135.48.149 already blocked

🕒 ⚡ Rate limit applied to 10.135.48.149: 10 req/min

➡➡ IP 10.135.48.149 already blocked

🕒 ⚡ Rate limit applied to 10.135.48.149: 10 req/min

Close

📶

HEALTHCARE

IoT Medical Device Network

TOTAL PACKETS28

THREATS28

🌱

AGRICULTURE

Smart Farming Sensors

TOTAL PACKETS0

THREATS0

🏙️

URBAN

Smart City Infrastructure

TOTAL PACKETS22

THREATS9

THREAT_MONITOR.exe ~ Live Feed

00:17:18[HEALTHCARE]BLOCKEDDDoS/Botnet Activity Detected (High traffic rate, High SYN count, High RST count, Suspicious packet timing, High packet volume)

00:17:16[HEALTHCARE]BLOCKEDDDoS/Botnet Activity Detected (High traffic rate, High SYN count, High RST count, Suspicious packet timing, High packet volume)

00:17:15[URBAN]ALLOWEDPacket analyzed

00:17:16[HEALTHCARE]BLOCKEDDDoS/Botnet Activity Detected (High traffic rate, High SYN count, High RST count, Suspicious packet timing, High packet volume)

00:17:08[URBAN]BLOCKEDMalicious Web Payload Detected (SQLi/XSS)

00:17:07[HEALTHCARE]QUARANTINEDoM Traffic Surge (DDoS). Prob: 1.00

00:17:05[HEALTHCARE]BLOCKEDDDoS/Botnet Activity Detected (High traffic rate, High SYN count, High RST count, Suspicious packet timing, High packet volume)

00:16:59[URBAN]ALLOWEDPacket analyzed

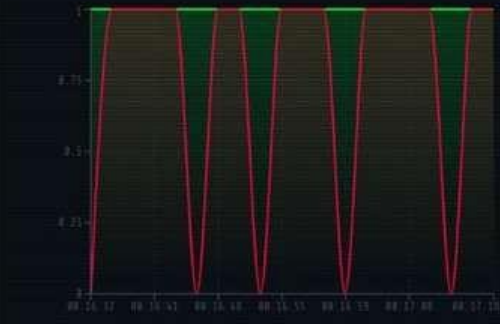
00:16:38[HEALTHCARE]BLOCKEDDDoS/Botnet Activity Detected (High traffic rate, High SYN count, High RST count, Suspicious packet timing, High packet volume)

00:16:16[HEALTHCARE]QUARANTINEDoM Traffic Surge (DDoS). Prob: 1.00

00:16:05[URBAN]BLOCKEDMalicious Web Payload Detected (SQLi/XSS)

📶 NETWORK ACTIVITY

TrafficThreats



50TOTAL PACKETS

13ALLOWED

37BLOCKED

74.0%THREAT RATE

IronDome .ai
Security Operations Center

ConnectedEvents: 0/minAlerts: 50Blocked: 10011:30:42
Thu, Aug 27, 2026

SearchCtrl K

Tanmay Loq Node
TANMAY-LOQ-nodeOnline

TelemetryTimelineIoT FleetResponseDropped 100

🌐 FLEET NETWORK

1 registered nodes across network

📶 PHYSICAL NODES

● TANMAY-LOQ

IP: 10.1.147.125Port: 5050

Sector: HEALTHCARE

ONLINE

📶 VIRTUAL IOT DEVICES (HEALTHCARE)

● pacemaker-01PACEMAKER

Integrity55%

● mr1-scanner-02MRI_SCANNER

Integrity55%

● infusion-pump-03INFUSION_PUMP

Integrity55%

50Total

17Warning

33Critical

⚠️ Critical Memory Usage

Memory usage exceeded 90% on Tanmay Loq Node. System may become unresponsive.

Tanmay Loq Node • 1s ago

✓ Ack✕

⚠️ ML: Network Shield🔍 AI

DDoS/Botnet Activity Detected (High SYN count)

Confidence: 65%

Unknown • 1s ago

✓ Ack✕

🕒 High Network Traffic (Data Exfiltration Risk)

Network traffic spike detected at 3525 KB/s. This could indicate data exfiltration or a flood attack.

✓ Ack All✕ Clear Ack'd

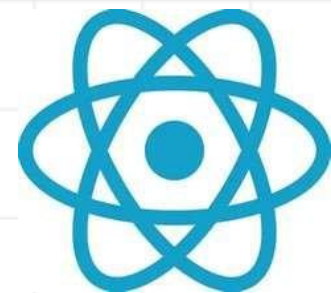
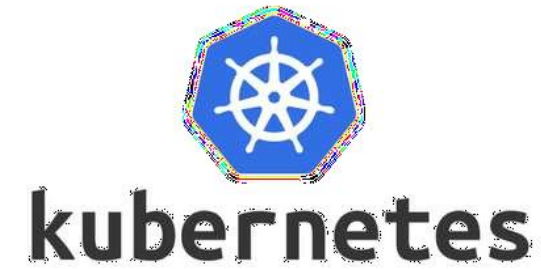
🛡️ Run Defense Playbook

1Total Nodes

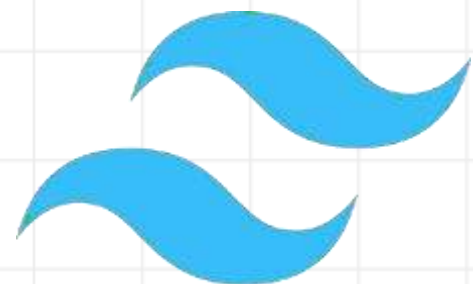
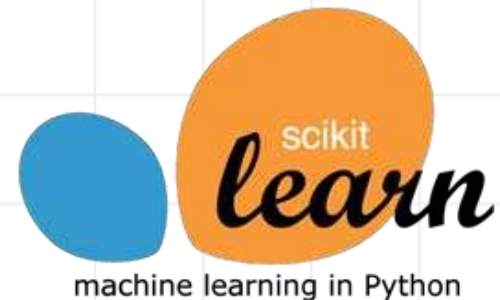
100%Health

ATTACK SIMULATION





TECHNOLOGIES USED



RESOURCES

DATASET LINKS :

- [HTTPS://WWW.KAGGLE.COM/DATASETS/PETERFRIEDRICH1/CICDARKNET2020-INTERNET-TRAFFIC](https://www.kaggle.com/datasets/peterfriedrich1/cicdarknet2020-internet-traffic)
- [HTTPS://DATA.MENDELEY.COM/DATASETS/2BW34GHT8B/2](https://data.mendeley.com/datasets/2bw34ght8b/2)
- [HTTPS://WWW.KAGGLE.COM/DATASETS/WISAM1985/IOT-AGRICULTURE-2024](https://www.kaggle.com/datasets/wisam1985/iot-agriculture-2024)
- [HTTPS://WWW.KAGGLE.COM/DATASETS/HIMADRI07/CICIOT2023/DATA](https://www.kaggle.com/datasets/himadri07/ciciot2023/data)
- [HTTPS://WWW.KAGGLE.COM/DATASETS/EVG3N1J/HTTPPARAMSDATASE](https://www.kaggle.com/datasets/evg3n1j/httpparamsdataset)